1. Abstract, research question, and contribution

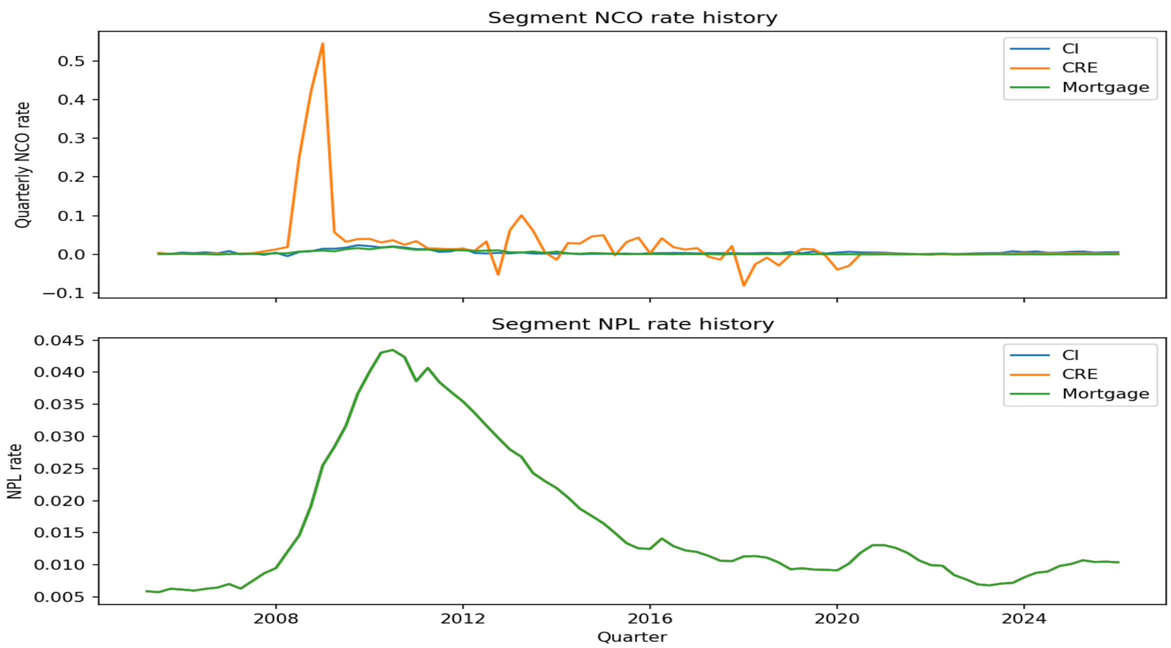
This reproducible public-data framework reconstructs segment-level quarterly net charge-off (NCO) rates for U.S. regional banks and translates the approved Federal Reserve 2026 scenarios into transparent loss-to-starting-Tier-1-capital results.

The generated panel contains 33 banks, 8,145 bank-segment-quarter observations, and 66 mapped raw fields. Results are descriptive conditional stress estimates, not failure predictions or causal effects.

2. Data, sample, and regulatory fields

The analysis unit is bank x loan segment x quarter. The panel uses FFIEC Call Report source snapshots, an effective-dated field mapping, documented merger flags, and FDIC noncurrent-loan controls.

Raw archives are excluded from version control; manifests retain official URLs, timestamps, and hashes. Missing values are retained as missing rather than converted to zero.



3. NCO reconstruction and quality assurance

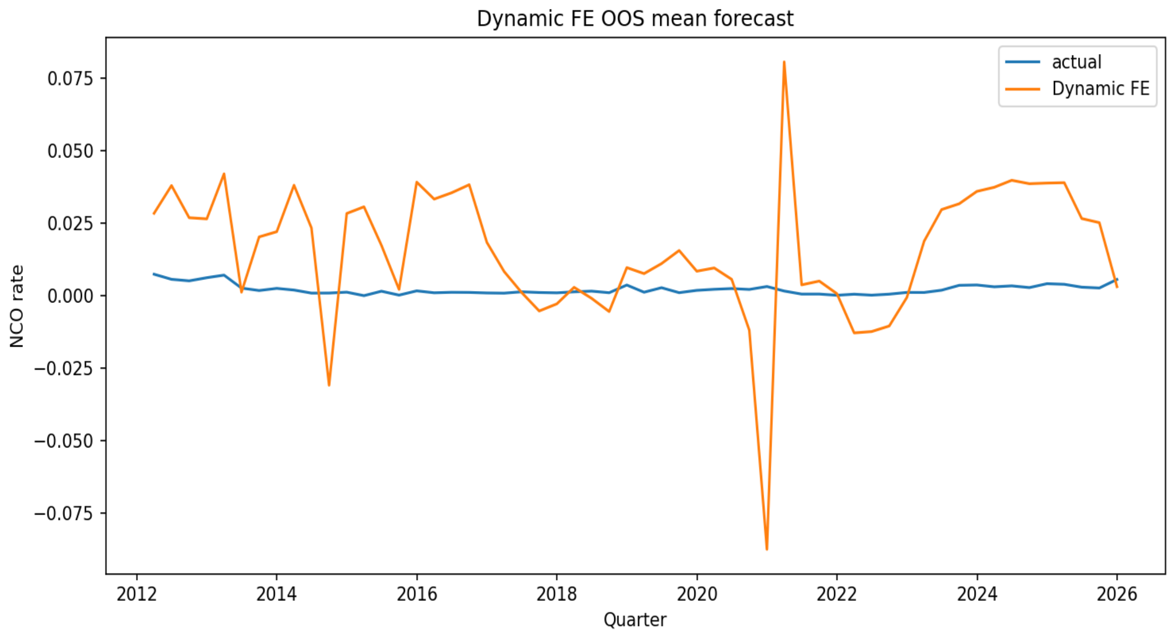
NCO is quarterly charge-offs less recoveries, divided by average segment exposure. YTD flows are quarterlyized before aggregation. The reproducibility audit retains the one documented REVIEW_REQUIRED source reconciliation item rather than recoding it.

Mortgage coverage is reconstructed for historical panel reporting, but no approved Mortgage stress model exists; stress tables therefore label mortgage attribution unavailable rather than zero.

4. Dynamic fixed effects and CRE interaction

The Dynamic-FE model is the approved mean model. The CRE interaction uses lagged CRE-to-Tier-1 measured at or before the forecast origin, following the documented timing repair. The analysis separates mechanical exposure-driven loss from any incremental loss-rate interaction.

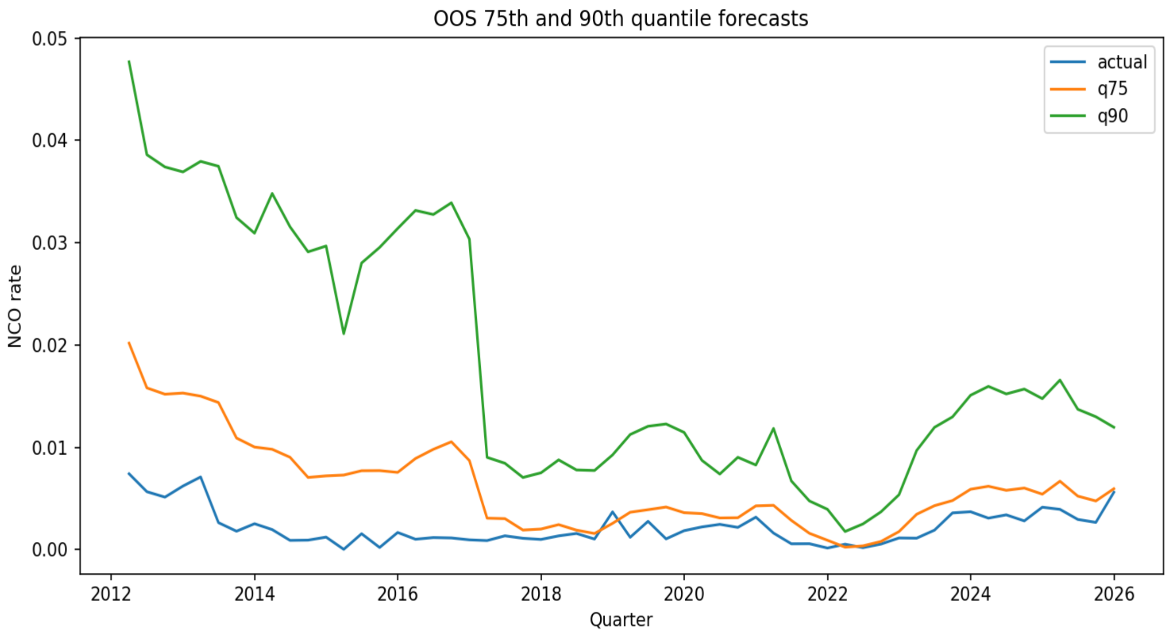
The committed pooled Dynamic-FE OOS comparison is reported in T3; the generated resume metric records an AR-relative RMSE change of 0.00%, not a pre-planned target.



5. Quantile and Bayesian model-risk evidence

Q0.50/Q0.75/Q0.90 forecasts are evaluated with pinball loss and pre-specified historical pseudo-stress windows. The Q0.90 CRE recursive path is not historically calibrated across GFC, COVID, and the 2022+ high-rate/CRE window; it remains a downstream tail sensitivity, not a validated forecast.

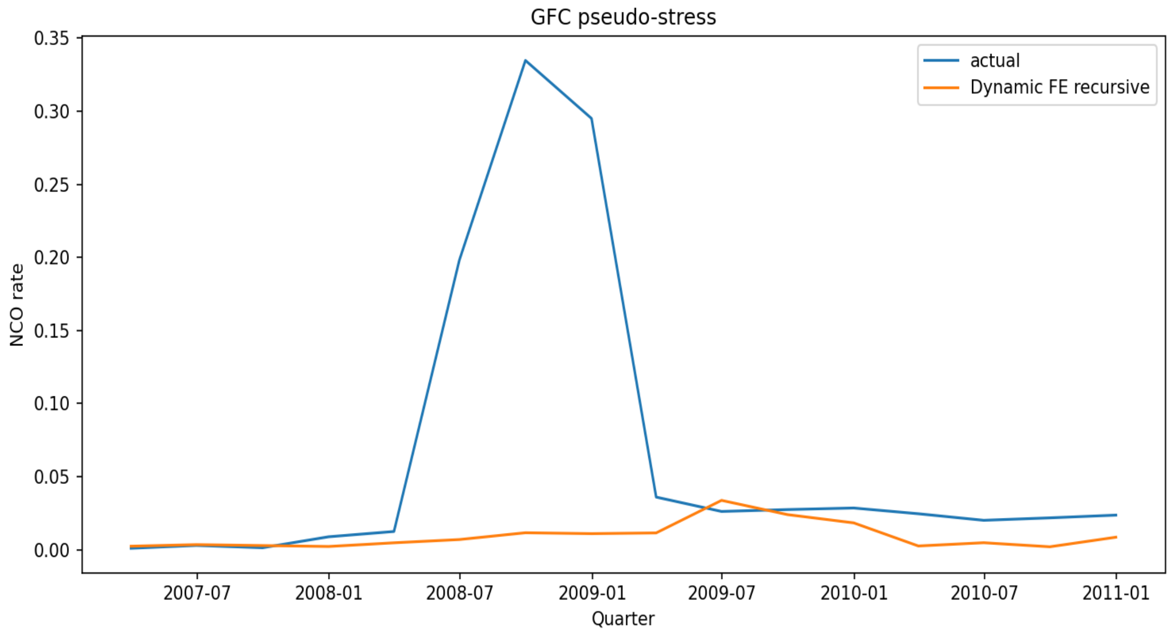
Bayesian posterior forecasts are unavailable under the recorded Batch 3 environment fallback. No sampled posterior result, interval, or performance claim is presented here.



6. OOS validation and historical pseudo-stress

All OOS windows preserve chronology and do not randomly shuffle observations. Pseudo-stress forecasts use realised historical macro paths, recursive NCO states, and frozen future bank controls.

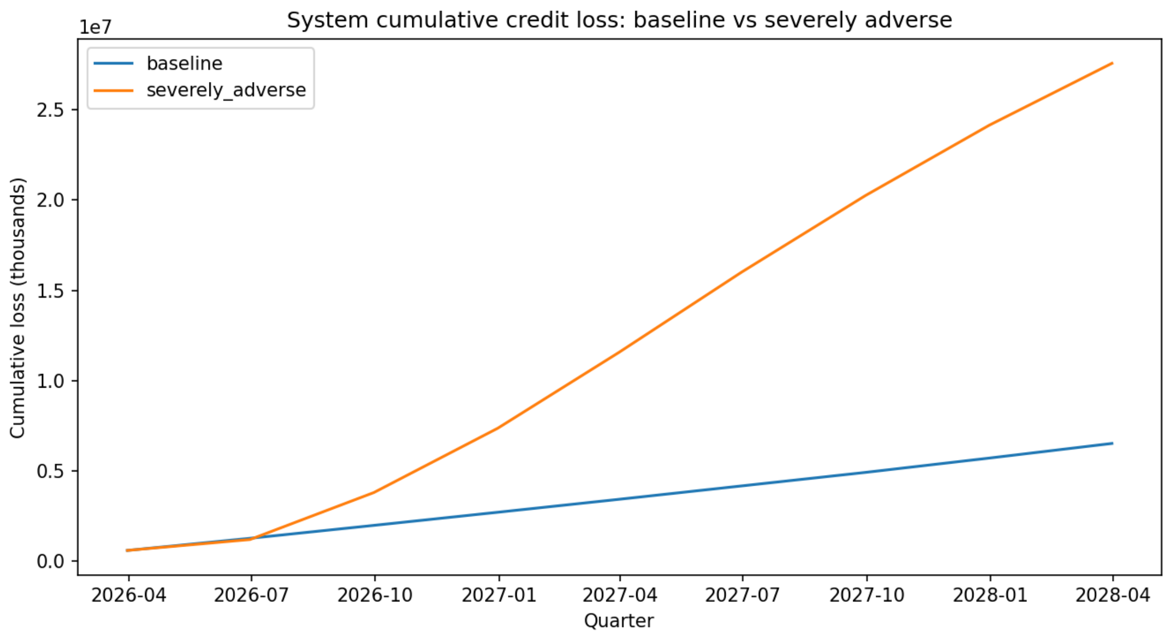
The OOS and pseudo-stress charts are generated from saved forecasts. Tail calibration failures are documented rather than optimized away after observing results.



7. Federal Reserve 2026 stress results

The project applies the official final baseline and severely adverse domestic paths, plus clearly labelled researcher sensitivity scenarios. Primary capital output is cumulative credit loss divided by starting Tier 1 capital, not a Federal Reserve CET1 projection.

The final stress universe is 14 banks with complete CRE and C&I; 2025Q4 states. The high-minus-low CRE-group mean capital-depletion difference is 28.41% under the Dynamic-FE severe scenario.



8. Model risk, robustness, and conformal fallback

Batch 5 implements a rolling residual-bootstrap interval fallback around OOS Dynamic-FE forecasts. It uses only earlier OOS residuals: an eight-quarter seed for the static comparator and a twenty-quarter adaptive calibration window. It is not presented as Bayesian or quantile conformal inference.

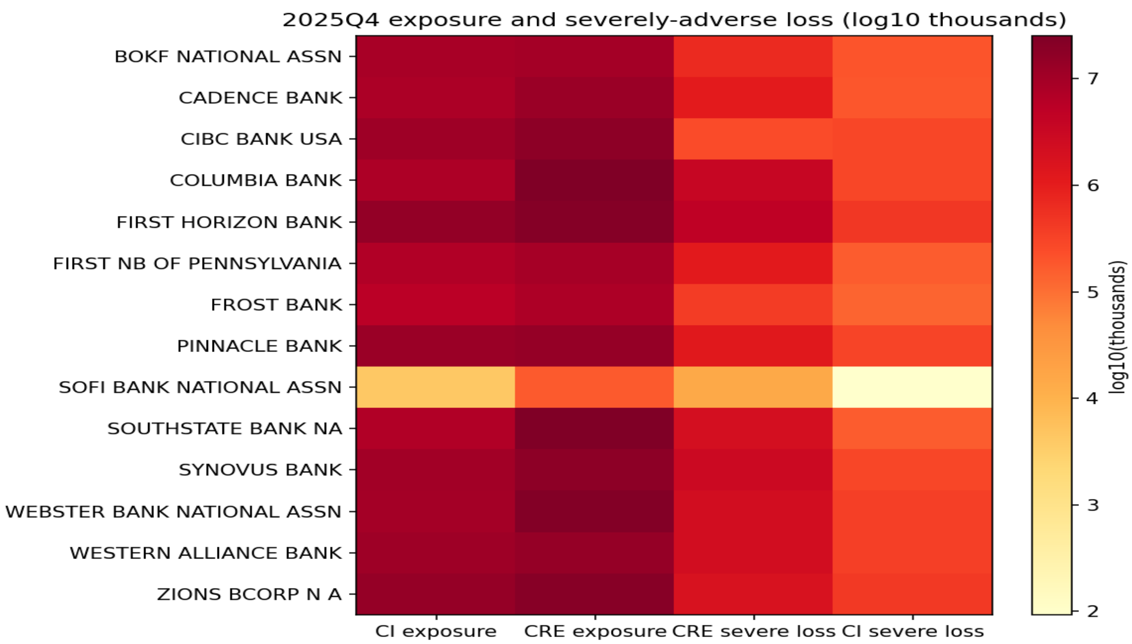
T5 reports static versus rolling residual-calibrated 90% interval coverage, width, proper Winkler score, and crisis underprediction counts by segment. In this run, rolling calibration increases crisis upper misses: CI 9 to 21; CRE 27 to 31. It therefore does not meet the desired crisis-underprediction reduction. The method targets calibration, not a guaranteed exact 90% result or a trading signal.

Method	Segment	Coverage	Width	Crisis upper misses
rolling_adaptive_residual_bootstrap	CI	94.4%	0.0254	21
rolling_adaptive_residual_bootstrap	CRE	92.1%	0.2133	31
static_residual_bootstrap	CI	96.9%	0.0373	9
static_residual_bootstrap	CRE	92.8%	0.2212	27

9. External validation and reproducibility

Market validation is deliberately not completed. No verified bank legal entity -> BHC/parent -> ticker mapping was available in this repository, and ambiguous name-to-ticker matching would violate the approved specification. No market-validation statistic is claimed.

The final audit checks artifacts, time ordering, effective-dated mapping fields, merger records, saved Bayesian diagnostics, non-zero missingness handling, and agreement between stress paths and reported stress totals.



10. Conclusion and limitations

The deliverable is a transparent public-data research pipeline with explicit model-risk evidence. It supports conditional portfolio stress analysis but does not reproduce confidential FR Y-14 supervisory models or make bank-failure predictions.

Carry-forward limitations: source/reconciliation caveats remain visible; macro final-vintage fallback variables are not real-time vintages; Bayesian posterior outputs are unavailable; recursive CRE Q0.90 is not historically calibrated; and Mortgage stress attribution is unavailable. Future work should verify a legal-entity parent/ticker mapping before external market validation.